AIMD-HF: A Minimal, Transparent, and Interactive Educational Implementation of Hartree–Fock Ab Initio Molecular Dynamics with Real-Time OpenGL Visualization

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Abstract

This repository contains a fully self-contained, educational implementation of real-time *Ab Initio Molecular Dynamics* (AIMD) using the Restricted Hartree–Fock (RHF) method with the minimal STO-3G basis set. The simulator supports hydrogen clusters (H_n), helium atoms (He_n), and protonated helium hydride systems (HeH^+). All electronic structure quantities are computed *on-the-fly*, allowing direct visualization of the coupling between quantum electronic structure and nuclear dynamics. The implementation prioritizes **conceptual clarity, transparency, and pedagogical value** over computational performance.

Introduction

AIMD-HF is an interactive educational simulator that combines real-time molecular dynamics with explicit quantum mechanical calculations. Unlike classical molecular dynamics based on empirical force fields, this program computes energies and forces directly from the electronic Schrödinger equation using the Hartree–Fock approximation at every time step.

This enables users to observe, in real time, how:

- electronic structure determines forces,
- forces drive nuclear motion,
- and molecular properties evolve dynamically.

The inclusion of He and HeH^+ systems allows exploration of weak interactions, charge asymmetry, and non-covalent behavior.

Key Features

- **Multi-system support:**
  - Hydrogen clusters (H_n)
  - Helium atoms (He_n)
  - Protonated helium hydride (HeH^+)
- Full Restricted Hartree–Fock (RHF) implementation with STO-3G basis
- Explicit computation of:
  - Overlap (S)
  - Kinetic energy (T)
  - Nuclear attraction (V_{ne})

  - Electron repulsion integrals (ERI)

- Self-consistent field (SCF) procedure with adaptive damping
- Nuclear forces via finite differences
- Velocity Verlet integrator
- Langevin thermostat
- Berendsen barostat
- Reflective boundary conditions
- Multi-threaded design (physics + rendering separation)

Real-Time Visualization and Analysis

- OpenGL-based 3D rendering
- Atoms as spheres and bonds as cylinders
- Real-time energy monitoring
- Dynamic Heads-Up Display (HUD)
- **Orbital visualization modes:**
  - Total electron density
  - HOMO visualization
  - LUMO visualization
- **Interactive educational panels (zero extra SCF cost):**
  - Molecular orbital diagram

  - Mulliken population analysis
  - Energy decomposition
  - Force vectors
  - Orbital volume rendering
  - Electric dipole panel
  - Statistics panel

- **Additional observables:**
  - Electric dipole moment (vector + magnitude)
  - HOMO–LUMO gap
  - Electronic virial ratio
  - Temperature and pressure

Keyboard Controls

Camera and Environment

Key	**Action**
W / S	Zoom in/out
A / D	Rotate camera (left/right)
Z / X	Rotate camera (up/down)
+ / -	Temperature ±50 K
B	Toggle background (dark/light)

Electronic Structure Visualization

Key	Action
V	Total electron density $\rho(\mathbf{r})$
H	HOMO orbital $\phi_{\mathrm{HOMO}}(\mathbf{r})$
L	LUMO orbital $\phi_{\mathrm{LUMO}}(\mathbf{r})$
O	Molecular orbital spectrum $\{\varepsilon_i\}$
M	Mulliken populations q_A
E	Energy decomposition $(T + V_{\mathrm{ne}} + V_{\mathrm{ee}} + V_{\mathrm{nn}})$
Y	Electric dipole moment $\boldsymbol{\mu}$
F	Force vectors $\mathbf{F}_A$
I	Statistics panel: step/time, total energy (Ha, eV), temperature, pressure, distance (bohr, Å), reference bond, HOMO–LUMO gap (Ha, eV), virial ratio $\frac{2\langle T\rangle}{\lvert\langle V\rangle\rvert}$

Simulation Control

Key	Action
P / SPACE	Pause / resume simulation
ENTER	Single MD step (when paused)
Q	Reset simulation (reproducible initial conditions)
ESC	Exit

Installation and Compilation

Linux

```
sudo apt install build-essential freeglut3-dev libglu1-mesa-dev
```

```
g++ aimd_hf.cpp -o aimd_hf -lGL -lGLU -lglut -lm -lpthread -O2
```

```
./aimd_hf
```

Windows (MSYS2)

```
pacman -S mingw-w64-x86_64-gcc
pacman -S mingw-w64-x86_64-freeglut

g++ aimd_hf.cpp -o aimd_hf -lfreeglut -lopengl32 -lglu32 -O2
```

Execution Parameters

```
./aimd_hf h (Default h2)
./aimd_hf h N
./aimd_hf he (Default he)
./aimd_hf he N
./aimd_hf heh+ (Default heh+)
./aimd_hf heh+ N
```

- N = number of atoms or pairs
- Default system: H_2
- Hydrogen systems require even N

Educational Value

This simulator enables direct exploration of:

- SCF convergence during dynamics

- Electronic structure–nuclear coupling
- Charge redistribution
- Orbital evolution (HOMO/LUMO)
- Energy decomposition
- Dipole formation
- Virial theorem behavior
- Quantum forces in real time
- Spatial structure of molecular orbitals

The goal is to transform abstract quantum concepts into **visible, dynamic phenomena**.

Supported Systems

- H_n: covalent bonding
- He_n: weak interactions
- HeH^+: ionic bonding

Design Philosophy

- Clarity over performance
- Explicit algorithms
- Minimal dependencies
- Strong pedagogical focus

Limitations

- $\mathcal{O}(N^4)$ scaling
- Finite-difference forces
- Small systems ($\sim$12 atoms)
- STO-3G basis only

Source Code

Available at:

`https://github.com/gallactuz/AIMD-HF`